

Supplementary Material: Supervision-Matched Selection of Paired RAG Repairs

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1 Fixed-action sensitivity

The primary analysis reallocates the global top- K actions within every question-cluster bootstrap draw. The following pre-specified sensitivity holds the observed action memberships fixed while reusing the same 20,000 draws and six-endpoint multiplicity adjustment.

Table 1: Fixed-action sensitivity. Values are percentage points.

Ordered comparison	Δ EM	Adjusted range	Δ Damage	Adjusted range	Joint rule
ROA-FULL – HGB+GbV _R	+0.0444	[-0.1333, +0.2278]	+0.0611	[+0.0000, +0.1278]	Not met
HGB+GbV _R – HGB-only _R	+0.0556	[-0.2000, +0.3111]	-0.0833	[-0.1722, +0.0000]	Not met
HGB+GbV _R – GbV-only _R	+0.2333	[+0.0389, +0.4278]	-0.0722	[-0.1389, -0.0222]	Met

2 Dataset breakdown

The cells below describe the one global allocation after partitioning by dataset. Each cell contains 6,000 traces. Actions, Recovery, Damage, and Net are counts; the EM/F1 differences from that cell’s Keep baseline are percentage points with the cell denominator. The cells do not reallocate 900 actions within each dataset and support no subgroup inference.

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
2wikimultihopqa	Keep	0	0	0	+0	+0.0000	+0.0000
2wikimultihopqa	HGB	292	136	12	+124	+2.0667	+2.6822
2wikimultihopqa	GbV	319	131	11	+120	+2.0000	+2.6189
2wikimultihopqa	ROA-FULL	298	134	8	+126	+2.1000	+2.8632
2wikimultihopqa	ROA-NOGBV	300	127	15	+112	+1.8667	+2.5163
2wikimultihopqa	HGB+GbV _R	297	129	4	+125	+2.0833	+2.8465
2wikimultihopqa	HGB-only _R	290	133	9	+124	+2.0667	+2.6644
2wikimultihopqa	GbV-only _R	318	132	10	+122	+2.0333	+2.6489
2wikimultihopqa	V2	313	139	10	+129	+2.1500	+2.8482
hotpotqa	Keep	0	0	0	+0	+0.0000	+0.0000
hotpotqa	HGB	323	132	16	+116	+1.9333	+2.6733
hotpotqa	GbV	303	98	10	+88	+1.4667	+2.1477
hotpotqa	ROA-FULL	315	122	10	+112	+1.8667	+2.6628
hotpotqa	ROA-NOGBV	316	121	17	+104	+1.7333	+2.4339
hotpotqa	HGB+GbV _R	309	121	7	+114	+1.9000	+2.7132
hotpotqa	HGB-only _R	286	119	12	+107	+1.7833	+2.4747
hotpotqa	GbV-only _R	300	99	10	+89	+1.4833	+2.1814
hotpotqa	V2	317	123	14	+109	+1.8167	+2.5511
musique	Keep	0	0	0	+0	+0.0000	+0.0000
musique	HGB	285	98	4	+94	+1.5667	+1.9603
musique	GbV	278	89	4	+85	+1.4167	+1.7499

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
musique	ROA-FULL	287	115	4	+111	+1.8500	+2.2084
musique	ROA-NOGBV	284	107	6	+101	+1.6833	+1.9843
musique	HGB+GbV _R	294	102	0	+102	+1.7000	+2.1124
musique	HGB-only _R	324	105	5	+100	+1.6667	+2.0959
musique	GbV-only _R	282	92	4	+88	+1.4667	+1.8109
musique	V2	270	99	4	+95	+1.5833	+1.9524

3 Retriever breakdown

The cells below describe the same global allocation after partitioning by retriever. Each cell contains 6,000 traces. Counts and percentage-point differences have the same meaning as above, using the retriever-cell denominator. These cells support no retriever-specific superiority claim.

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
bm25	Keep	0	0	0	+0	+0.0000	+0.0000
bm25	HGB	386	149	15	+134	+2.2333	+2.9645
bm25	GbV	318	111	7	+104	+1.7333	+2.3527
bm25	ROA-FULL	291	119	4	+115	+1.9167	+2.5157
bm25	ROA-NOGBV	296	117	8	+109	+1.8167	+2.2911
bm25	HGB+GbV _R	300	118	2	+116	+1.9333	+2.5663
bm25	HGB-only _R	212	100	2	+98	+1.6333	+2.0095
bm25	GbV-only _R	295	109	5	+104	+1.7333	+2.3324
bm25	V2	391	154	10	+144	+2.4000	+3.0984
dense	Keep	0	0	0	+0	+0.0000	+0.0000
dense	HGB	282	130	8	+122	+2.0333	+2.5925
dense	GbV	276	101	6	+95	+1.5833	+2.0890
dense	ROA-FULL	305	129	5	+124	+2.0667	+2.7886
dense	ROA-NOGBV	307	122	15	+107	+1.7833	+2.4558
dense	HGB+GbV _R	312	125	3	+122	+2.0333	+2.7335
dense	HGB-only _R	395	150	13	+137	+2.2833	+3.0715
dense	GbV-only _R	305	112	7	+105	+1.7500	+2.3119
dense	V2	271	113	9	+104	+1.7333	+2.3713
hybrid	Keep	0	0	0	+0	+0.0000	+0.0000
hybrid	HGB	232	87	9	+78	+1.3000	+1.7588
hybrid	GbV	306	106	12	+94	+1.5667	+2.0747
hybrid	ROA-FULL	304	123	13	+110	+1.8333	+2.4302
hybrid	ROA-NOGBV	297	116	15	+101	+1.6833	+2.1875
hybrid	HGB+GbV _R	288	109	6	+103	+1.7167	+2.3724
hybrid	HGB-only _R	293	107	11	+96	+1.6000	+2.1540
hybrid	GbV-only _R	300	102	12	+90	+1.5000	+1.9970
hybrid	V2	238	94	9	+85	+1.4167	+1.8821

4 Policy construction details

All five supervised heads use 3,600 fit questions and 900 disjoint calibration questions, with three retriever siblings retained within a question group. Numeric imputation and standardization use fit data only. Missing-value indicators and one-hot retriever indicators are unscaled. The base classifiers and Platt calibrators use fixed logistic-regression settings and are never refit after fresh outcomes are opened. Raw HGB, raw GbV, and V2 are included for descriptive context but are not supervision-matched attribution controls.

5 Frozen historical HGB score

Identity and target. HGB means the historical *histogram-based gradient-boosting* pairwise selector implemented with `HistGradientBoostingClassifier`. It is a frozen upstream estimator and is not the HGB-only_R head fitted in the current experiment. Its historical development universe contains 7,200 labeled traces from 4,800 question groups and 1,539 available pair-feature records. Only 601 informative traces from 518 question groups enter the final fit. For normalized exact-match outcomes $c_i \in \{0, 1\}$ of answer a_i , the historical target is

$$y = \begin{cases} 1, & (c_0, c_1) = (0, 1), \\ 0, & (c_0, c_1) = (1, 0). \end{cases}$$

Pairs with $(c_0, c_1) = (0, 0)$ or $(1, 1)$ have no preference label and are excluded from HGB fitting. Thus HGB learns which member of an already acquired answer pair is correct when exactly one member is correct; it does not learn the current experiment’s full-population Recovery target from all eligible traces.

Likelihood cells and branch features. For $i, j \in \{0, 1\}$, L_{ij} is the mean teacher-forced token log probability of the fixed answer a_i when the reader is conditioned on the question and evidence set E_j . Hence L_{00} and L_{11} are the two own-evidence likelihoods, while L_{01} and L_{10} score each answer under the other branch’s evidence. For branch i , define the eight-component vector

$$\phi_i = (\ell_i^{\text{own}}, \ell_i^{\text{cross}}, \ell_i^{\text{own}} - \ell_i^{\text{cross}}, n_i, u_i^+, u_i^-, s_i^+, s_i^-),$$

where $\ell_0^{\text{own}} = L_{00}$, $\ell_0^{\text{cross}} = L_{01}$, $\ell_1^{\text{own}} = L_{11}$, and $\ell_1^{\text{cross}} = L_{10}$. Here n_i is answer token length; u_i^+ and u_i^- are multiset token F1 between a_i and the added and removed document text; and s_i^+ and s_i^- are `SequenceMatcher` ratios between normalized a_i and those texts. The last two historical field names contain “semantic compatibility,” but their frozen implementation is normalized string-sequence similarity rather than a neural semantic model.

Let $\delta = \phi_1 - \phi_0$. The five pair-context variables are

$$z = (o_E, d_S, n_E, t_A, s_A),$$

where o_E is the Jaccard overlap of document IDs in E_0 and E_1 ; d_S is added-document retrieval score minus removed-document retrieval score; n_E is one minus the largest token-set Jaccard similarity between the added document and any document in E_0 ; t_A is token F1 between a_0 and a_1 ; and s_A is their saved BGE answer-pair similarity. The HGB input is

$$x = [\delta; \delta \otimes z] \in \mathbb{R}^{8+8 \times 5} = \mathbb{R}^{48}.$$

This compact construction expands to eight branch differences and forty difference–context interactions. Dataset and retriever identities are absent from the historical HGB input.

State symmetry, model, and score direction. The fit duplicates each historical example as (x, y) and $(-x, 1 - y)$. The fixed classifier uses `max_iter=300`, `max_leaf_nodes=15`, learning rate 0.08, L2 regularization 1.0, and final-fit seed 20261834; no feature scaling or hyperparameter search is applied to this final HGB fit. If $p_H(1 | x)$ is its class-one probability, the stored repair-preference score is

$$h(x) = \frac{1}{2}\{p_H(1 | x) + 1 - p_H(1 | -x)\}.$$

Consequently $h(-x) = 1 - h(x)$ up to numerical precision, and a larger $h(x)$ favors replacing a_0 with a_1 . Raw HGB ranks eligible traces directly by $h(x)$. HGB-only_R is a separate current L2-logistic Recovery head whose inputs are $h(x)$, one missingness flag, and three retriever indicators; HGB+GbV_R adds the paired GbV margin and its flag. The two current heads share the current fit/calibration recipe and never refit HGB.

Source and provenance boundary. The authenticated historical definitions are `src/mars/state_symmetric.py` (SHA-256 3724b5ac77722b70cabdf2589379d5584942f34ec81f3f5a04f88e0665f8f717) and the model binding in `src/mars/full_experiment.py` (SHA-256 65713509ef6c8d094bb44ad2c47b0238ceade607616d3e729b31c56096326d50). The original saved HGB artifact is 639,652 bytes with SHA-256 9245170f855435b5603b04013bd4fbab79a75d2761853a012ab9f3259600bf6e. A source-assembled replay reproduced all 1,539 HGB scores with zero numerical error. Its serialized file was not byte-identical because the saved bin mapper recorded one CPU-thread metadata value instead of six; all observed learned arrays and tree fields matched. The original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable, so this evidence authenticates the saved computation and reconstructed fit, not the literal historical execution event.

6 Answer metrics

Normalization lowercases text, removes ASCII punctuation and the English articles `a`, `an`, and `the`, and collapses whitespace. EM is one if the normalized prediction equals any allowed reference. Token F1 uses multiset token overlap and takes the maximum over allowed references. Unequal answers involving `yes`, `no`, or `noanswer` receive zero F1; an empty token sequence scores one only against another empty sequence. The method-blind independent metric implementation applies these same conventions. No new reference evaluation is performed for manuscript preparation.

7 Data and historical-development scope

The evaluation source frames are HotpotQA validation, 2WikiMultiHopQA development, and MuSiQue training. The last label describes the public benchmark split, not membership in the current selector’s fit set. A prospectively fixed hash order selected 2,000 questions per dataset after excluding the recorded 18,600-question union; the result is a deterministic frame-scoped cohort, not a probability sample or semantic-decontamination certificate.

The seven historical upstream estimators have reconstructed membership of 601 fit traces / 518 question groups within a 7,200-trace / 4,800-question historical development universe. Those 4,800 groups and the 3,000 earlier V2 questions belong to the exclusion union and have zero recorded dataset/ID overlap with the current five-head development cohort. Only the five current heads share the 4,500-question, 3,600-fit / 900-calibration recipe. Upstream learning histories are part of the fixed signal dependency, so matched current heads do not imply equal total historical label budgets for every signal.

8 Provenance and stopped extensions

Saved estimator bytes, reconstructed historical membership, numerical replay, pre-label scoring witnesses, action memberships, outcome mapping, aggregate analysis, and independent validators are preserved under hashes. A locally preserved pre-ROA-named package contains a byte-identical copy of the complete historical training manifest, but its timestamp has no independent external certification. Seven original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable. These disclosures do not exclude unknown unlogged activity. The Qwen/BGE replay covers a prospectively selected 180 of 18,000 traces; downstream saved neural witnesses are fully checked, but this is not a second complete neural-forward run.

A Phi reader extension acquired artifacts but failed its fixed semantic validation because 6,573 of 9,120 BF16 query vectors exceeded the 10^{-3} tolerance, with maximum deviation 0.0041683525. It was excluded before policy fitting and fresh effect analysis. A Mistral extension stopped at protocol review before engineering or model execution. Neither is a failed reader-effect experiment, and neither supports a robustness claim.

9 Workload disclosure

Stage timers cover distinct receipt boundaries and must not be summed: C2 1,096.562 s; C3 45,216.887 s; base scoring 5,337.573 s; GbV scoring 3,523.022 s; allocation 157.658 s. Separate saved C4 peaks of 8,153,970,176 and 2,956,706,304 bytes are also non-additive. The bounded replay separately contains 540 generation receipts, 518,966 prompt tokens, and 8,441 generated tokens/forwards. It is a reproduction witness, not an independent workload estimate. Standalone policy and per-generation latency, a uniform end-to-end timer, canonical C2/C3 peaks, a simultaneous whole-system peak, C2/C3 BGE token totals, generation input-token touches per autoregressive forward, FLOPs, and energy were not measured. Missing values are not zero; prompt or output-token totals are not FLOP estimates.

10 Reproduction boundary

The anonymous aggregate candidate permits deterministic checking of published numbers, interval signs, joint decisions, and file hashes. It excludes raw benchmark text, model weights, local paths, per-example outcomes, and private immutable ledgers. This supports aggregate reporting verification. It does not yet support public end-to-end regeneration of retrieval, answers, neural scores, or upstream training.